

STOVETOP MONITOR

CONCEPT DOCUMENT

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OVERVIEW.

This document presents the concepts for a Stovetop Monitor. The Stovetop Monitor monitors the state of a stovetop and alarms if it senses that a burner is turned on for too long.

PROBLEM.

A stovetop burner that is accidentally left on presents a fire hazard. Busy/distracted people as well as elderly people who are becoming forgetful can accidentally leave a stovetop burner on, unattended. In the worst case, this can lead to a fire hazard and burn down the house as well as risking lives. In less than the worst case, it can lead to damage to the stove and/or cookware.

There are few, if any, stovetop/ovens on the market that provide remote access via an app. In general, interconnected stoves have apps that can monitor and control an oven but do not have a similar capability vis-a-vis the stovetop burners. These apps are aimed at remote control of baking in the oven and it is assumed that cooking on the stovetop will always be attended.

Modern stovetop/ovens tend not to have mechanical controls. Rather, they are electronically controlled via an embedded microcontroller. This means that the status of the stovetop burners cannot be monitored nor can they be controlled remotely by hacking into electro-mechanical switches. An external solution - one that hopefully does not modify the stovetop/oven in any way - needs to be developed.

STOVETOP MONITOR CONCEPT.

One possible solution to stovetop monitoring is to use a non-contact IR temperature sensor that "looks down" on the stovetop and measures the temperature within its field of view (FOV). Either a single, wide angle sensor might work, or else an IR camera sensor might be needed. The former is cheaper and simpler. For example, an MLX-60914 sensor has a 90 degree FOV and can report both ambient temperature and the average temperature within its FOV:

- <https://www.melexis.com/en/documents/documentation/datasheets/datasheet-mlx90614>

The ability of such a sensor to accurately detect when a burner is on needs experimentation. If such a wide area sensor does not work, perhaps an IR camera sensor can be used;; e.g.:

- <https://www.adafruit.com/product/3538?srltid=AfmBOorppc3F6FdUb-WMJUKhOnkLOmnM3Nvxe29fIR-7lpoXQprGYatK>

The basic idea is to have a non-contact IR sensor looking down at the stovetop and compare the ambient temperature with the temperature of the stovetop/area on the stovetop. Readings would be taken often – once every several seconds at most. Temperature difference thresholds would be set to indicate a burner is on medium - high, on warm - low, or off. If the temperature difference readings indicate that a burner has been turned on, a timer is set. An alarm is generated if the stovetop remains on, in a sustained state, for a pre-designated period of time; e.g.:

- If the temperature reading indicates a burner is on medium - high heat, alarm after e.g. ½ hour.
- If the temperature reading indicates a burner is on warm - low heat, alarm after e.g. 1 hour.
- Reset the time if the temperature reading indicates that no burner is on or if a temperature threshold is passed (NOTE: the assumption here is that changes to a burner setting must be manual, implying that the stove is attended at that time).

Both a local alarm and a remote alarm should be generated. A local alarm would beep a buzzer and/or flash an LED. A remote alarm would send an SMS text or Pushover alert to a pre-configured mobile phone.

In the event that an IR camera array sensor is used, each zone in the array would have its own set of thresholds and timer.

Every installation and every type of stove will need to have its own set of calibrations for detection of a burner being on.

A mobile phone app may be part of this project. The app will allow the user to read out the current sensor reading(s) at any time and will allow the user to remotely reset alarms.

REQUIREMENTS.

Sensing Requirements.

- The Stovetop_sensor shall continuously measure ambient and stovetop temperatures.

- The Stovetop_sensor shall have one or more thresholds for stovetop temperature.
- If the stovetop temperature minus the ambient temperature exceeds a threshold, the sensor shall determine the amount of time during which the threshold is exceeded.
- If there are multiple thresholds, the Stovetop_sensor shall determine the amount of time that has elapsed between threshold crossings.
- If the elapsed time since the last threshold crossing exceeds a preset criteria, an alarm shall be generated. Different thresholds may be set for different combinations of threshold crossings, based upon different assumed cooking scenarios.
- Alarms shall persist until reset by the user.

Stove Requirements.

- The Stovetop_sensor shall be able to monitor the stovetop from a position above and away from the stove.
- The Stovetop_sensor shall process and monitor temperature readings from an electric glass top stove.
- The Stovetop_sensor shall optionally process and monitor temperature readings from a gas burner stove.
- The Stovetop_sensor shall optionally process and monitor temperature readings from an electric direct burner stove.
- Note: the Stovetop_sensor is not required to work with an induction stove; these stoves often include their own shutdown sensors and circuitry.

Alarming Requirements.

- The Stovetop_sensor shall sound a local alarm by flashing an LED and beeping a buzzer.
- The Stovetop_sensor shall have a button that, if pressed, resets the alarm and clears out the alarm timer.
- The Stovetop_sensor shall remotely alarm the user via either an SMS text or a Pushover alert to a configured mobile phone.

App Requirements.

- The Stovetop_sensor may have an Android (optionally, iPhone) app.
- The app shall allow the user to display the current ambient temperature and the current stovetop temperature reading that are being processed by the Stovetop_sensor.
- The app shall display the current stovetop status (warming, cooking, etc.) and the alarm timer status (elapsed time since the last threshold was passed).
- The app shall permit the user to reset the alarm remotely (via the app).

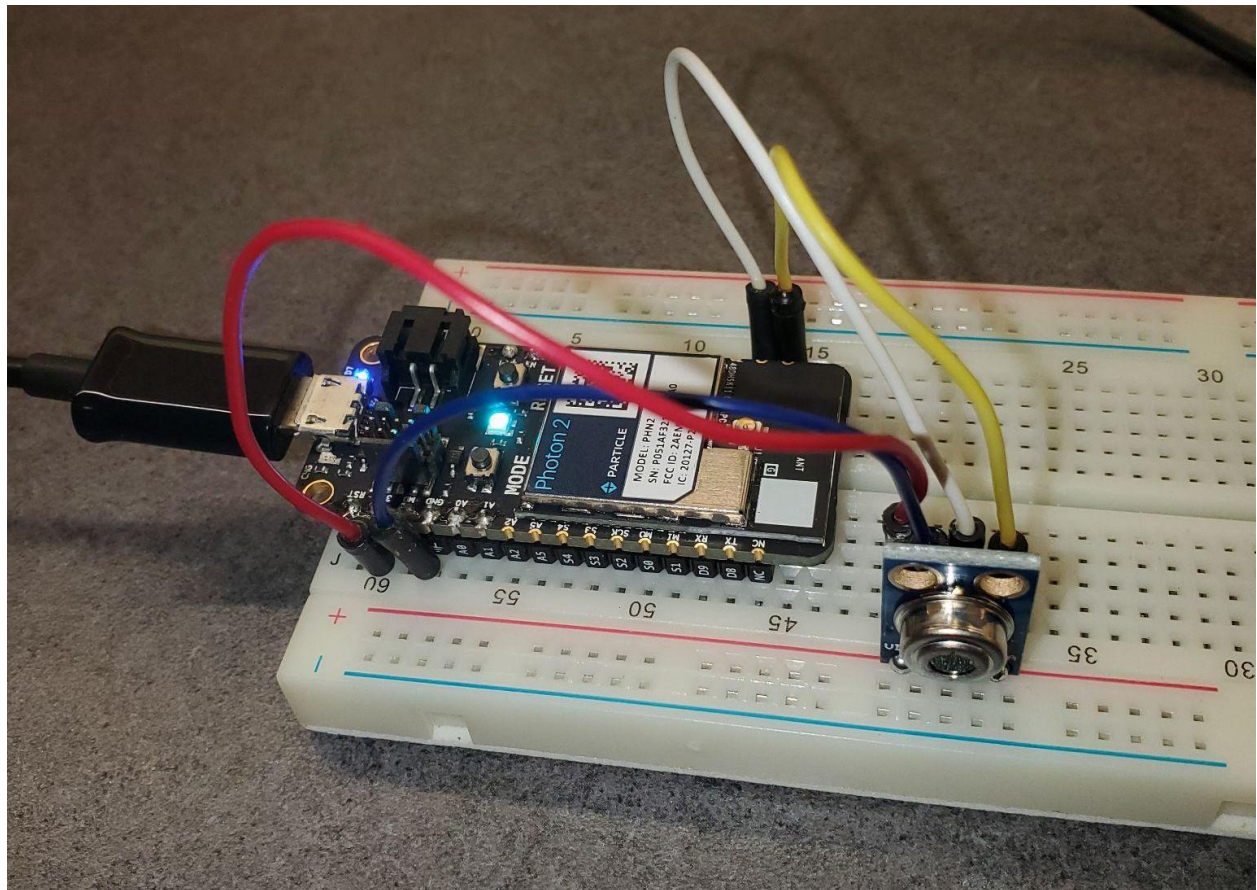
Device Requirements.

- The device shall have a button that can be pressed to silence the alarms.
- The device shall have LEDs that indicate the internal state of the sensor: warming is detected; cooking is detected; alarm condition is detected.

EXPERIMENTAL PLAN.

It is currently not known whether remote IR sensing will work. Experiments are planned using an MLX90614 sensor module

(https://www.amazon.com/dp/B0B63K5V7T?ref=ppx_yo2ov_dt_b_fed_asin_title) and a Particle Photon 2.



The Particle Photon 2 will query the sensor every 15 seconds and report ambient and object temperatures to a serial monitor. The sensor will be positioned above and off to the side of the stovetop under test. Each of the burners will, in turn, be turned on to various heat levels and the temperature monitored to see if a reliable threshold can be obtained for each burner and for various heat levels. The tests will be performed first with bare burners and then repeated under actual cooking conditions.

The initial tests will be performed on a glass top electric stove. Additional experimental tests may be performed on other types of stoves, if available.

TEST RESULTS.

Tests were performed as planned. The results are recorded in a series of files, one per test scenario. The tests performed under controlled conditions on “naked” burners are recorded here:

https://drive.google.com/drive/u/0/folders/1eQ8ShkG1vRneR8z-zXsf0qDWCEcLc3_J

The data recorded from actual cooking events are recorded here:

<https://drive.google.com/drive/u/0/folders/1i1JrBY8-6Tzk80zNIIKa8FqDaMydmla>

These test results were graphed and analysed on spreadsheets. A summary of this analysis is recorded here:

<https://docs.google.com/spreadsheets/d/19OL7LbVIh1ShGyvxy2oRAUm9VLhCmWPMqeghsToV7wA/edit?gid=304464798#gid=304464798>

Some significant findings from these tests are:

- There is an issue with the sensor's I2C interface that sometimes sends a false result of 1899.59 deg F. These false readings must be detected and ignored in any operational code.
- When taking data every 15 seconds, burner warmup and cooldown is pretty much monotonic; however there are a few instances where there are small spikes in the data.
- When a burner is up to the preset temperature, the stovetop controls cycle the burner on and off, resulting in “spikey” data.
- Different burners are different sizes and at different distances and angles from the sensor. On the particular test stove, it is possible to set thresholds that distinguish when all burners are off, when at least one burner is on a low setting (warming) and when at least one burner is on a higher setting (cooking). However, there is too much overlap between burners and temperature settings in the test data to reliably set additional thresholds.
- The sensor reports both ambient temperature (the sensor body temperature) and the IR measured temperature. As one or more burners heat up, the ambient temperature of the sensor itself rises, but the IR temperature usually rises to a greater degree. This is as expected. The conclusion is that the best measure of what is happening on the stovetop is to use the difference between IR temperature and ambient temperature as the measurement criteria.

- When a pot or pan is placed on a burner, the IR temperature measurement is usually lower than for a naked burner. The difference may be small or large, depending upon the pot/pan material, the pot/pan color, and the contents of the pot/pan. When the pot/pan is removed from the burner, the IR temperature will jump up (perhaps significantly), even if the burner has been turned off.
- When all burners are off and the ambient temperature is room temperature, the IR temperature reading is usually at or below the ambient temperature reading. When the furthest and smallest burner is turned on to its lowest setting, the IR temperature rises to more than 1.5 degrees F above ambient. This is the worst case; with other burners on the lowest setting, the difference between IR and ambient is much higher. It is therefore possible to determine, reliably, when any burner is turned on.
- Testing of cooking on multiple burners as well as baking/broiling in the oven have not been performed, but these scenarios should be taken into account when devising an operational algorithm.

OPERATIONAL ALGORITHM.

The state diagram below represents an algorithm that satisfies the experimental data to date.

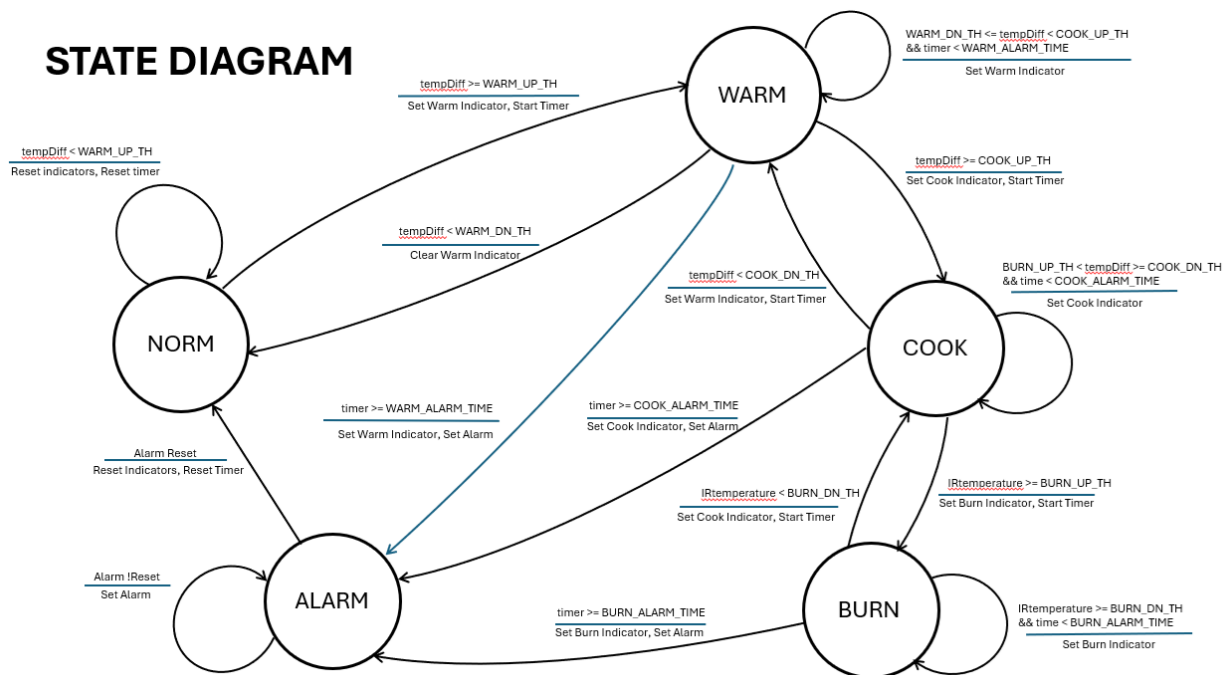


Figure 1. Stovetop Monitor State Diagram.

The following states of the stovetop are represented in figure 1:

- NORM. All burners are turned off - the “normal” state.

- WARM. One or more burners are turned on, to low or medium heat.
- COOK. One or more burners are on medium to high heat.
- BURN. The indicated temperature of the stovetop is higher than expected for normal cooking.
- ALARM. When the stovetop is in the WARM, COOK, or BURN state longer than expected (exceeding a state-specific timeout), the ALARM state is entered. The ALARM must be reset (by user action - either pressing a button on the electronics or on a cell phone app) in order to exit this state and return to the OFF state.

Transitions between NORM, WARM, COOK, and BURN states are based upon temperature thresholds. The ambient temperature reported by the sensor is subtracted from the IR temperature reported by the sensor. This value is compared with a threshold to determine if a state transition is to occur.

Separate thresholds are defined for heating up and for cooling down. This provides an hysteresis that prevents “noise” on the readings from causing invalid state transitions. “Noise” on the readings can occur because of the burner cycling, breezes near the sensor, and other phenomena.

Transitions between WARM, COOK and BURN states and the ALARM state are time driven. If the sensor remains in a state for too long a time, the ALARM state is entered. Any transition between NORM, WARM, COOK and BURN states resets the timer to begin a new timeout in the newly entered state.

Transition out of the ALARM state requires a manual action on the part of the user, and always returns the sensor to the NORM state. It is assumed that the user has taken some action to ensure that all burners are off, or that the cooking situation is otherwise normal and safe, before resetting from an alarm. If cooking is still taking place, the algorithm will proceed through the state diagram again, automatically.

The thresholds that cause state transitions are defined as follows:

- WARM_UP_TH(RESHOLD). If this threshold is crossed while in the NORM state, the WARM state is entered. The timer is reset and begins accumulating time in the WARM state. The value of 1.5 degrees F will be used for the glasstop electric stove, based upon experimentally derived data.
- COOK_UP_TH(RESHOLD). If this threshold is crossed while in the WARM state, the COOK state is entered. The timer is reset and begins accumulating time in the COOK state. The value of 7 degrees F will be used for the glasstop electric stove, based upon experimentally derived data.
- BURN_UP_TH(RESHOLD). If this threshold is crossed while in the COOK state, the BURN state is entered. The timer is reset and begins accumulating time in the BURN

state. The threshold is 130 degrees F on the IR sensor, regardless of ambient temperature, on the glasstop electric stove, based upon experimentally derived data.

- WARM_DN_TH(RESHOLD). If this threshold is crossed while in the WARM state, the NORM state is entered, indicating that the stovetop burners are all off. The value of 1.0 degrees F will be used for the glasstop electric stove, based upon experimentally derived data.
- COOK_DN_TH(RESHOLD). If this threshold is crossed while in the COOK state, the WARM state is entered, indicating that the stovetop is cooling down. The timer is reset and begins accumulating time in the WARM state. The value of 6 degrees F will be used for the glasstop electric stove, based upon experimentally derived data.
- BURN_DN_TH(RESHOLD). If this threshold is crossed while in the BURN state, the COOK state is entered. The timer is reset and begins accumulating time in the COOK state. The threshold is 125 degrees F on the IR sensor, regardless of ambient temperature, on the glasstop electric stove, based upon experimentally derived data,

The time-based transitions to the ALARM state are based upon experimental data and assumptions based upon various cooking scenarios:

- WARM_ALARM_TIME. If the stovetop is in the WARM state for longer than this threshold, it is assumed that one or more burners have been left on unintentionally and the ALARM state is entered. This threshold is nominally 65 minutes, based upon experimental data and assumed cooking scenarios.
- COOK_ALARM_TIME. If the stovetop is in the COOK state for longer than this threshold, it is assumed that one or more burners have been left on unintentionally and the ALARM state is entered. This threshold is nominally 40 minutes, based upon experimental data and assumed cooking scenarios
- BURN_ALARM_TIME. If the stovetop is in the BURN state for longer than this threshold, it is assumed that one or more burners have been left on unintentionally and the ALARM state is entered. This threshold is nominally 5 minutes, based upon experimental data and assumed cooking scenarios.

The IR sensor data will be read out every 15 seconds. This interval was used in the stovetop experiments and seems to be a good compromise between reading too often and waiting too long between readings. Reading too often results in “noisy” data, while reading at too long an interval could result in delayed reaction to alarming conditions.

The experiments sometimes produced erroneous readings of 1899.59 degrees F, on either or both of the ambient temperature and the IR temperature. These are spurious readings that

must be ignored. Anytime that such a spurious reading is obtained, the previously read value will be used.

PHYSICAL DESIGN CONCEPT.

A 3D printed enclosure will mount the sensor and its associated electronics underneath a cabinet that is above but off to the side of the stovetop. The enclosure will have a 45 degree sloping face so that the sensor is looking down at the center of the stovetop. The sensor, indicators, and electronics will all be in the enclosure.

The enclosure housing the electronics should be removable for servicing. The concept is to have a mounting bracket that affixes to the under lip of the cabinet and/or to the underside bottom of the cabinet using wood screws. The enclosure containing all electronic components should slide into the bracket and lock into place, preferably without the need for any retaining screws.

The electronics enclosure should mount the following components on its front face:

- GREEN LED
- YELLOW LED
- RED MOMENTARY BACKLIT PUSHBUTTON

These visual components will indicate the current state of the sensor as follows:

- GREEN LED only is lit: the electronics is powered on and in the NORM state.
- YELLOW LED only is lit: the electronics are powered on and in the WARM state.
- YELLOW LED only is *flashing*: the electronics are powered on and in the COOK state.
- RED PUSHBUTTON backlight only lit: the electronics are powered on and in the BURN state.
- RED PUSHBUTTON backlight only is *flashing*: the electronics are powered on and in the ALARM state. An internal buzzer will also be beeping when in the ALARM state.
- All LEDs and backlights are off: the unit is not powered.

The RFID Station printed circuit board will accommodate these components, and will be used on this project:

- A Particle Photon 2 will be the processor. It fits into the “Argon” connector of this PCB.
- The 74HCT125 level shifter will be used to drive the LEDs and buzzer.
- A piezo buzzer will be used on the pcb.
- The IR sensor module will be connected to the I2C bus connector on the PCB, The I2C pullup resistors on the PCB **will not** be soldered onto the board, as the sensor has its own pullups.

- The READY connector will drive the GREEN LED
- The ACCEPT connector will drive the YELLOW LED. The pushbutton inputs on this connector will not be used.
- The REJECT connector will connect the RED BACKLIT PUSHBUTTON to the PCB.
- Connections for the LCD display and the LCD contrast potentiometer will not be used on this project.
- 5 volt USB power will be supplied to the electronics via the usb connector on the Photon 2.

APP DESIGN CONCEPT.

Use of a cellphone app is optional for this project. A user who is at home when a stovetop alarm goes off will be alerted by the flashing red light and the beeping buzzer. The user can clear the alarm by pressing the backlit pushbutton on the sensor enclosure.

A user who is not at home will be notified of a stovetop alarm via an SMS text message and/or Pushover notification on their mobile phone. The remote user need not reset the alarm remotely, as there is nobody at home to deal with it anyway. The alarm simply notifies the remote user to take some action to have someone physically attend to the alarm.

On the other hand, it will be helpful for the remote user to have a cellphone app available that ties into the stovetop alarm system. This will allow the user to remotely reset the alarm. The proposed app will display the following information to the user:

- The current state of the stovetop sensor (NORM, WARM, COOK, BURN, ALARM).
- The most current IR and ambient temperature readings from the sensor.
- The current time in the current state of WARM, COOK, BURN, or ALARM (time = 0 for NORM).

The app will use Particle's REST API to communicate this information. GET will be used to obtain the above information from global variables in the Photon 2 code that are exposed to the Particle Cloud by Particle.variable(). POST will be used to call a cloud function on the Photon 2 (Particle.function()) that resets the alarm. In the absence of the app, a remote user can access the same functionality via the Particle Console.

The app will be written in MIT App Inventor 2 and built for Android devices (.apk file). The app may optionally be built and deployed for Apple (iOS) devices. The latter will require an Apple developer account and may only be deployed via the Apple App Store.

OTHER DESIGN CONSIDERATIONS.

The temperature thresholds and alarm timeout values that are listed above are based upon very limited testing and a lot of assumptions about cooking scenarios that need to be monitored. It is highly likely that these values will need to be tweaked in the future, based upon real world experience, even for the specific stovetop used for the experimental testing done to date. Different types of stovetops may require very different settings.

All temperature and timeout thresholds need to be clearly defined in `#DEFINE` statements, either at the top of the main code or in a separate `#INCLUDE` file.

Serial port monitoring of the sensor data was used for the stovetop experiments and this capability should be preserved in the deployed firmware. In this way, a laptop can be used to record data from new cooking scenarios in order to fine tune the thresholds as more experience is gained.

A WORD ABOUT FALSE POSITIVES.

The algorithm embodied in the state diagram of figure 1 is designed to favor false positives over false negatives. Getting false alarms under normal conditions is annoying. Missing an alarm under abnormal conditions could result in fires and appliance damage.

Given the expectation of false positives, it might be useful for a remote user to have some means of verifying a mobile phone alarm. The user will want to know whether it was a false alarm or if they need someone to go into the house immediately to deal with a real, dangerous condition.

At the moment, it appears that the only way to verify a stovetop alarm is to actually view the stovetop; specifically, to view the stovetop control panel. An inexpensive webcam that is mounted on a cabinet near the stovetop is one way to provide realtime remote stovetop status. The use of such a webcam is recommended, but purely optional. Time and use of the stovetop sensor will tell how valuable this extra verification will be.

The notion of adding a webcam begs the question “why not use image processing on the webcam image to generate alarms in the first place”? This is technically feasible for any specific stovetop and webcam position combination. However, it is not a general solution. The proposed stovetop monitor approach is general, but prone to false positives, as noted above.